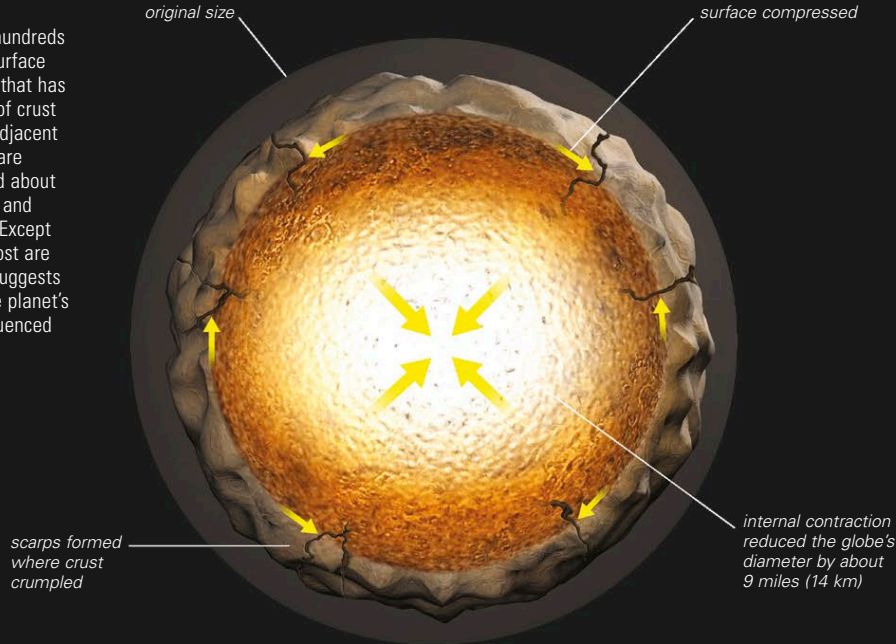


GLOBAL CONTRACTION
Escarpments traceable for hundreds of miles across Mercury's surface attest to global contraction that has caused the edges of tracts of crust to become thrust over the adjacent terrain. The biggest scarps are 1–2 miles (2–3 km) high and about 1,000 miles (1,500 km) long and date back billions of years. Except for those near the poles, most are oriented north-south. This suggests that the tidal slowing of the planet's rate of spin might have influenced their origin.



GEOGRAPHY

Mercury is an airless and heavily cratered world. Most of its surface consists of vast sheets of lava. The older the lavas, the more scarring has been caused by impact craters. The volcanic eruptions that supplied the lava waned to a trickle about 3 billion years ago, but the rate of impact cratering had waned by then, too, so that the youngest lava surfaces have relatively few craters and are described as “smooth plains.” Most plains on Mercury (planitia in Latin) are named after Mercury (the Roman messenger god) in various languages, whereas craters are named after famous artists, authors, and musicians. Raphael, Shakespeare, and Mozart each have a home on this planet. Scarps (rupes in Latin) are named after ships used on voyages of discovery.

MERCURY MAP
The entire globe was mapped for the first time by NASA's MESSENGER craft and is shown here in black-and-white images at 90-degree intervals.

MISSIONS TO MERCURY

Only two spacecraft have visited Mercury, and a third is on the way.

1974–1975 MARINER 10 (USA) Mariner 10 was placed in an orbit around the Sun that had exactly twice the period of Mercury's solar orbit. It flew past Mercury three times in 1974–1975, before its fuel supply ran out, but saw only the same half of the planet each time. It discovered the planet's unexpected magnetic field and sent back our first close-up views.

2011–2015 MESSENGER (USA) After making three flybys of Mercury, MESSENGER became the first spacecraft to orbit the planet, which it did for more than 4 years before crashing into the surface. It had a more sophisticated suite of instruments than Mariner 10 and carried a ceramic sunshade to protect itself from the Sun's heat. To avoid being close to Mercury's hot dayside surface for long periods, MESSENGER was placed into a strongly elliptical orbit.



2025 BepiColombo (ESA/JAXA)
Launched in 2018, BepiColombo is due to achieve orbit around Mercury in 2025. A larger European orbiter will concentrate on the planet's surface, and a smaller Japanese orbiter will concentrate on its magnetic field.

